

Week 07 - Class Assignment

JAWABAN

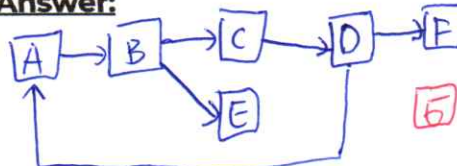
#1: Normalization

Determine the relational candidate key(s), non-key attribute, normal form, and the reason for the following relations. Draw a FD diagram first for each of the following relations.

a. R(A,B,C,D,E,F,G)

$A \rightarrow B$
 $B \rightarrow CE$
 $C \rightarrow D$
 $D \rightarrow AF$

Answer:



C. Key: $(A, \underline{B})$, $(B, \underline{E})$, $(C, \underline{D})$, $(D, \underline{F})$

NKA: E, F
 (PFD) (PFD)

$\therefore$ 1nf, karena NKA E dan F merupakan Partly Functional Dependency terhadap Key. / membutuhkan sebagian key

b. R(A,B,C)

$A \rightarrow B$
 $B \rightarrow A$

Answer:



Candidate Key: $(A, \underline{C})$, $(B, \underline{C})$

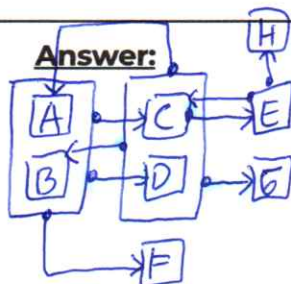
NKA: —

$\therefore$ 3nf, karena determinan A dan B bukan key.

c. R(A,B,C,D,E,F,G,H)

$AB \rightarrow CDF$
 $CD \rightarrow ABG$
 $C \rightarrow E$
 $E \rightarrow CH$

Answer:



C. Key: $(A, \underline{B})$, $(C, \underline{D})$, $(D, \underline{E})$

NKA: F, G, H
 (FFD) (FFD) (PFD)

$\therefore$ 1nf, karena NKA H hanya membutuhkan sebagian key

$AB \rightarrow C$ $E \rightarrow C$
 $AB \rightarrow D$ $E \rightarrow H$
 $AB \rightarrow F$
 $CD \rightarrow A$
 $CD \rightarrow B$
 $CD \rightarrow G$
 $C \rightarrow E$

d. R(A,B,C,D,E)

AB → CD
BD → AE

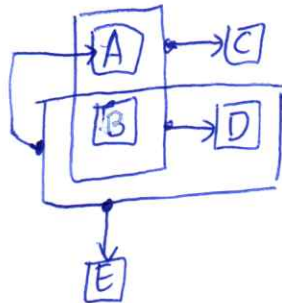
AB → C

AB → D

BD → A

BD → E

Answer:



Cand. key: (A, B), (B, D)

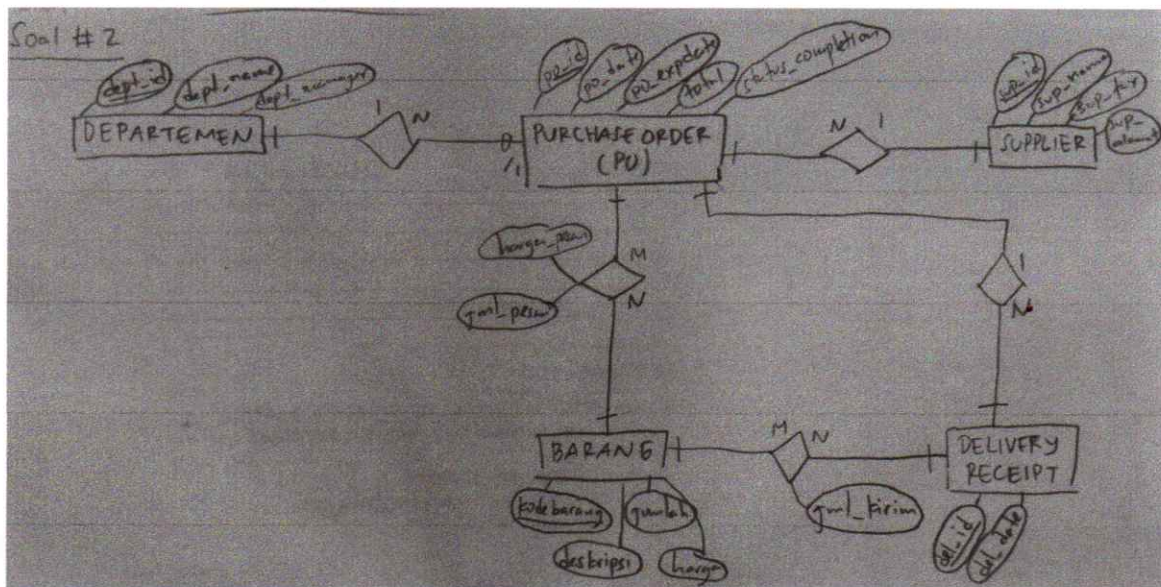
NKA: C, E

(FFD) (FFD)

∴ BCNF, karena semua determinant adl key.

#2: Relational Model

Convert this ER Model into a Relational Model. You have to list all relations, including column, primary key, and foreign key.



DEPARTEMEN (dept_id^{*}, dept_name, dept_managers)
 PURCHASE_ORDER (po_id^{*}, po_date, po_exp_date, total, status_completion, sup_id, dept_id)
 BARANG (kode_barang^{*}, deskripsi, jumlah, harga)
 SUPPLIER (sup_id^{*}, sup_nama, sup_fax, sup_alamat)
 DELIVERY_RECEIPT (del_id^{*}, del_date, po_id)
 PO_BARANG (po_id, kode_barang^{*}, harga_pesanan, jml_pesanan)
 DR_BARANG (del_id, kode_barang^{*}, jml_kirim)

 ^{*} = primary key
 red = foreign key